

Calculus Quiz Practice Paper (University Level)

Total Marks: 10 | Suggested Time: 40 minutes

Topics Covered

- Inequalities (Linear, Quadratic, Rational, Absolute Value)
- Functions (Types, Absolute Value, Graphs, Linear, Identity, Constant, Quadratic)
- Greatest Integer Function & Lowest Integer Function
- Composition of Functions & Combined Functions
- Limits
- Discontinuous / Piecewise Functions
- Continuous Functions

Instructions: Attempt all questions. Show complete working. Use exact values unless stated otherwise.

Section A: Inequalities (4 Marks)

Q1. Solve the inequality and express the solution set in interval notation: $3x - 5 < 2x + 7$

Q2. Solve the quadratic inequality: $x^2 - 5x + 6 \geq 0$

Q3. Solve the rational inequality: $(x + 2)/(x - 3) \leq 1$

Q4. Solve the absolute value inequality: $|2x - 3| > 5$

Section B: Functions & Graphs (4 Marks)

Q5. Let $f(x) = |x - 2|$. Sketch the graph and write $f(x)$ as a piecewise function.

Q6. Determine whether the function is one-to-one and/or onto: $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2 + 1$

Q7. Let $f(x) = 2x + 1$ and $g(x) = x^2 - 4$. (a) Find $(f \circ g)(x)$ (b) Find $(g \circ f)(x)$ (c) Evaluate $(f \circ g)(-1)$

Q8. Let $h(x) = \sqrt{x - 1}$ and $k(x) = 1/(x + 2)$. Find the domain of $(h \circ k)(x)$.

Section C: Limits, Continuity & Special Functions (2 Marks)

Q9. Evaluate the limit: $\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2)$

Q10. A function is defined as: $f(x) = \begin{cases} ax + 1, & x < 2 \\ x^2 + b, & x \geq 2 \end{cases}$ Find values of a and b such that $f(x)$ is continuous at $x = 2$.

Extra Challenge Questions (Not Included in 10 Marks) – For Practice

EC1. Let $[x]$ denote the greatest integer function. Solve: $[x] + [2x] = 7$

EC2. Let $\lceil x \rceil$ denote the least integer function. Solve: $\lceil x \rceil = 3$ and express the solution as an interval.

EC3. Determine the points of discontinuity of: $f(x) = (x^2 - 1)/(x - 1)$

EC4. Evaluate: $\lim_{x \rightarrow 0} (|x|/x)$ and discuss whether the limit exists.